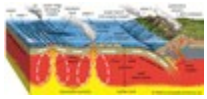
Time line of the development of the theory of plate tectonics

Significant events in the development of the theory of plate tectonics are summarized in the table.

Time line of the development of the theory of plate tectonics

	year	event
	1596	Flemish mapmaker Abraham Ortelius noted that the coastlines of the continents appear to fit together. He suggested that the continents were once joined and that the Americas were "torn away" from Europe and Africa.
	1912	German meteorologist and geophysicist Alfred Wegener proposed that the continents were once joined in a supercontinent called Pangea. Wegener believed that Pangea's constituent portions moved thousands of miles apart over long periods of geologic time, a phenomenon he called "continental displacement" (now known as continental drift). Until the 1950s and '60s, however, his idea was rejected by most geologists because he could not describe the driving forces behind continental drift.
	1929	British geologist Arthur Holmes proposed that convection in the mantle is the force driving continental drift. Although his ideas were not taken seriously at the time, Holmes's mantle convection hypothesis later gained support.
	1950s	Oceanographic vessels mapping the ocean floor provided data on the topographic features of the ocean basin, leading to the discovery of mid-ocean ridges. These underwater mountain ranges encircling the planet form as Earth's plates separate.

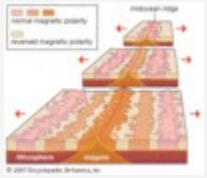
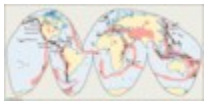
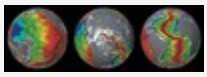
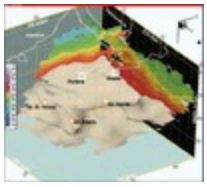
	1960	American geophysicist Harry H. Hess developed the idea that oceanic crust forms along mid-ocean ridges and spreads out laterally away from the ridges. The following year, geophysicist Robert S. Dietz named the phenomenon seafloor spreading. Hess and Dietz's work played a pivotal role in the development of the modern theory of plate tectonics.
	1963	British geologists Frederick J. Vine and Drummond H. Matthews—as well as Canadian geophysicist Laurence W. Morley, who worked independently of the others—postulated that new crust would have a magnetization aligned with Earth's geomagnetic field. They noted that this would appear over geologic time as bands of crust that exhibit alternating patterns of magnetic polarity. The later identification of such patterns of magnetic striping provided additional evidence that Earth's plates separate at mid-ocean ridges.
	mid-1960s	A global network of sensors designed to detect hydroacoustic signals was installed to monitor compliance with the Nuclear Test-Ban Treaty of 1963. The sensors also recorded earthquake activity. Scientists later found that earthquakes and volcanic activity occur almost exclusively at the edges of tectonic plates.
	1968	The vessel *Glomar Challenger* set sail on an exploration of the mid-ocean ridge between South America and Africa. Core samples obtained from drilling revealed that rocks close to mid-ocean ridges are younger than rocks that are farther away from the ridges.
	mid-1970s	Scientists created three-dimensional images of Earth's interior by combining information from many earthquakes using an approach similar to computed tomography (CT) scanning. This technique, now known as seismic tomography, enables scientists to investigate the dynamic processes in the deep interior of Earth.

Plate tectonics and the geologic past

The extent to which plate tectonics has influenced Earth's evolution through geologic time depends on when the process started. This is a matter of ongoing debate among geologists. The principal problem is that almost all oceanic crust older than about 200 million years has been obliterated by subduction. Although thick sequences of marine sedimentary rocks up to 3.5 billion years old imply that oceanic environments did exist early in Earth's history, virtually none of the oceanic crust that underlay these sediments has been preserved. Despite these disadvantages, there is enough fragmentary evidence to suggest that plate tectonic processes similar to those of today extend back in time at least as far as the Early Proterozoic Era, some 2.5 billion to 1.6 billion years ago.

plate tectonics. (2012). In *Encyclopædia Britannica*. Retrieved from http://www.britannica.com/EBchecked/topic/463912/plate-tectonics/261582/Unanswered-questions